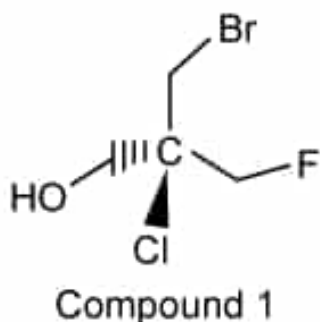


When the compound shown below undergoes an S_N2 reaction with hydroxide, which of the following compounds will most likely form as the major product?

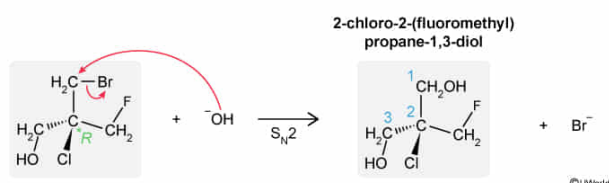


- ✓ ☒ A. 2-chloro-2-(fluoromethyl)propane-1,3-diol [32%]
☐ B. 2-(bromomethyl)-2-chloropropane-1,3-diol [12%]
☐ C. (2*R*)-3-bromo-2-(fluoromethyl)propane-1,2-diol [27%]
☐ D. (2*S*)-3-bromo-2-(fluoromethyl)propane-1,2-diol [27%]

Correct

32% answered correctly

Explanation:



Substitution reactions require a good **nucleophile** and a **good leaving group** on a suitable substrate. **S_N2 reactions** occur when a **nucleophile attacks** an electrophile, causing the **leaving group to leave** in the **same step**. These reactions take place more readily on the **less-substituted carbon** because they are less **sterically hindered**, allowing the nucleophile easier access to attack the electrophilic substrate.

Bromide and fluoride in Compound 1 are both **primary alkyl halides**, which are generally good substrates for S_N2 reactions. However, **halide leaving group ability increases** as the **size of the halide ion increases** because the increased ion surface area stabilizes the negative charge on the departing ion. The small size of fluoride makes it a **poor leaving group**, and therefore the attack by the hydroxide nucleophile occurs at the bromide site. Substitution of the bromide by the hydroxyl group forms a second $-CH_2OH$ group bonded to the central carbon, converting the compound into a diol and resulting in an **achiral molecule** (ie, a carbon atom with two identical substituents that is not a stereocenter).

To name the product molecule, the carbons of the longest continuous chain must be numbered such that the **highest priority functional group** (ie, both alcohol groups) are part of the main chain. This chain contains three carbons with the alcohol groups found on carbons 1 and 3, so the molecule is a propane-1,3-diol. The chloro– and fluoromethyl– substituents are then listed alphabetically with the number of the carbon to which they are bonded. This yields the name **2-chloro-2-(fluoromethyl)propane-1,3-diol**.

(Choice B) This would be the name of the compound if fluorine acted as the leaving group. However, fluorine is a poor leaving group, so this reaction is unfavorable and cannot form the major product.

(Choices C and D) These compounds could result if chloride were the leaving group. However, the chloride atom is tertiary, and therefore relatively unreactive. If **chloride** were to act as a leaving group, the *backside attack* associated with S_N2 reactions would cause a **stereocenter inversion**, changing the molecule to an *S* rather than an *R* configuration.

Educational objective:

S_N2 reactions are substitution reactions that occur when a nucleophile attacks the electrophile and causes the leaving group to leave in the same step. These reactions occur on the less-substituted carbon, and the halide leaving group ability increases as the size of the halide ion increases.

Time spent:0 sec

Subject:

Organic Chemistry

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System:

Introduction to Organic Chemistry